

UTOPIA ELECTRONIC DEVELOPMENT SPECIFICATIONS

Last updated 2/10/25

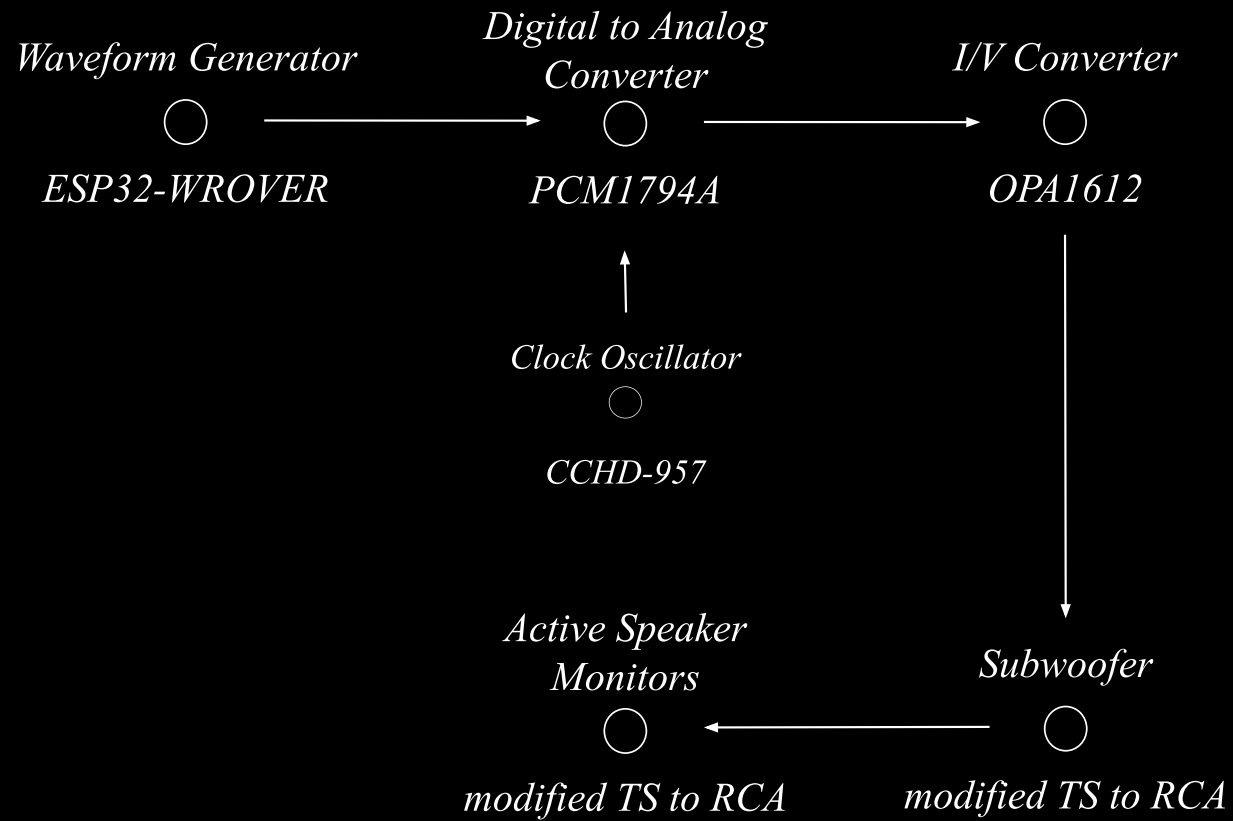
PROJECT OBJECTIVES

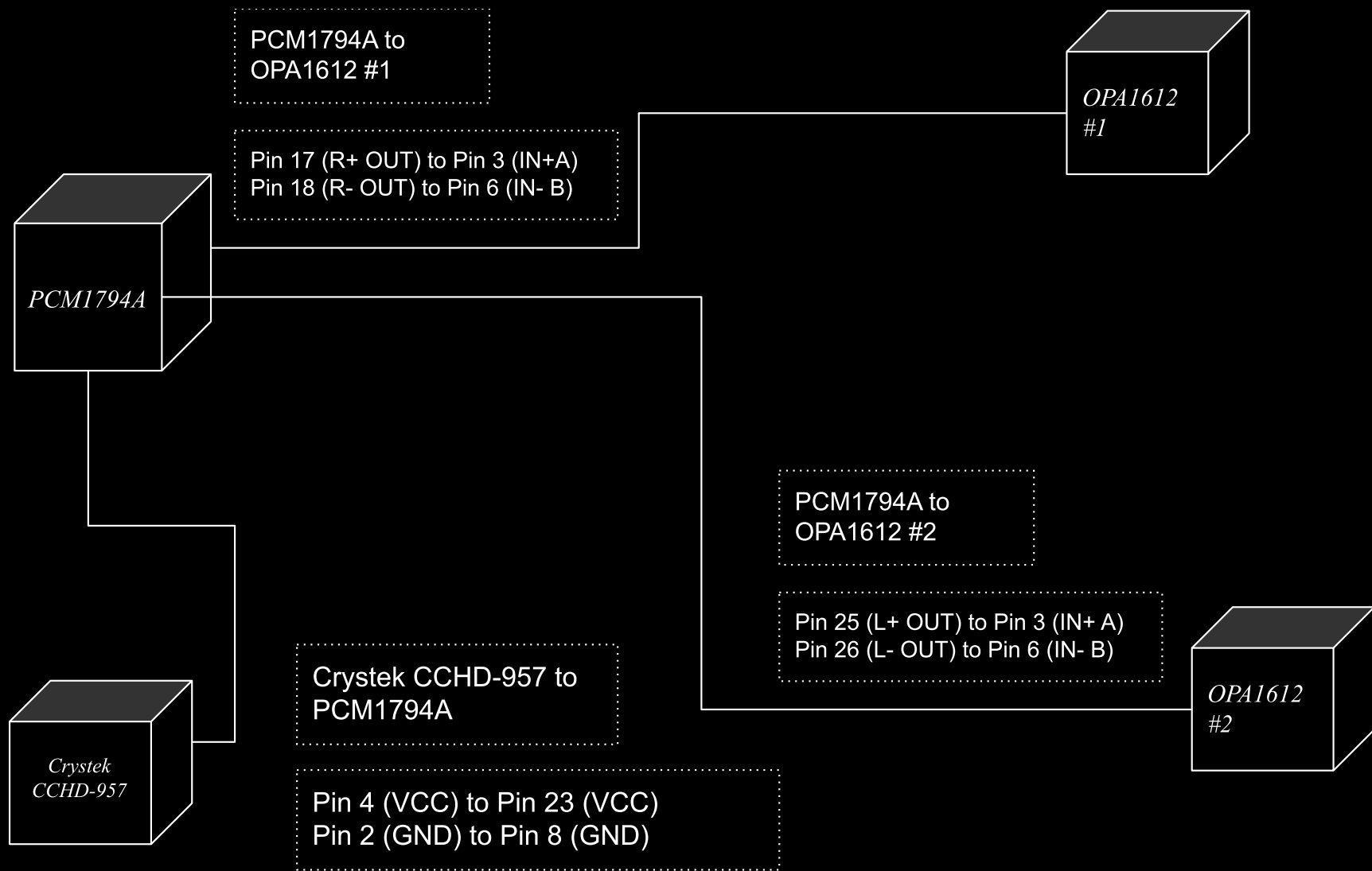
- Design a system of **ESP-32 microcontrollers** programmed to perform **waveform synthesis** and **audio effects** manipulated by **data** from **proximity sensors**.
- Integrate an **oscilloscope visual** for monitoring the performance of sound.
- Integrate **an external sound system** for the hardware synthesizer to output sound.

DEVELOPMENT FLOW

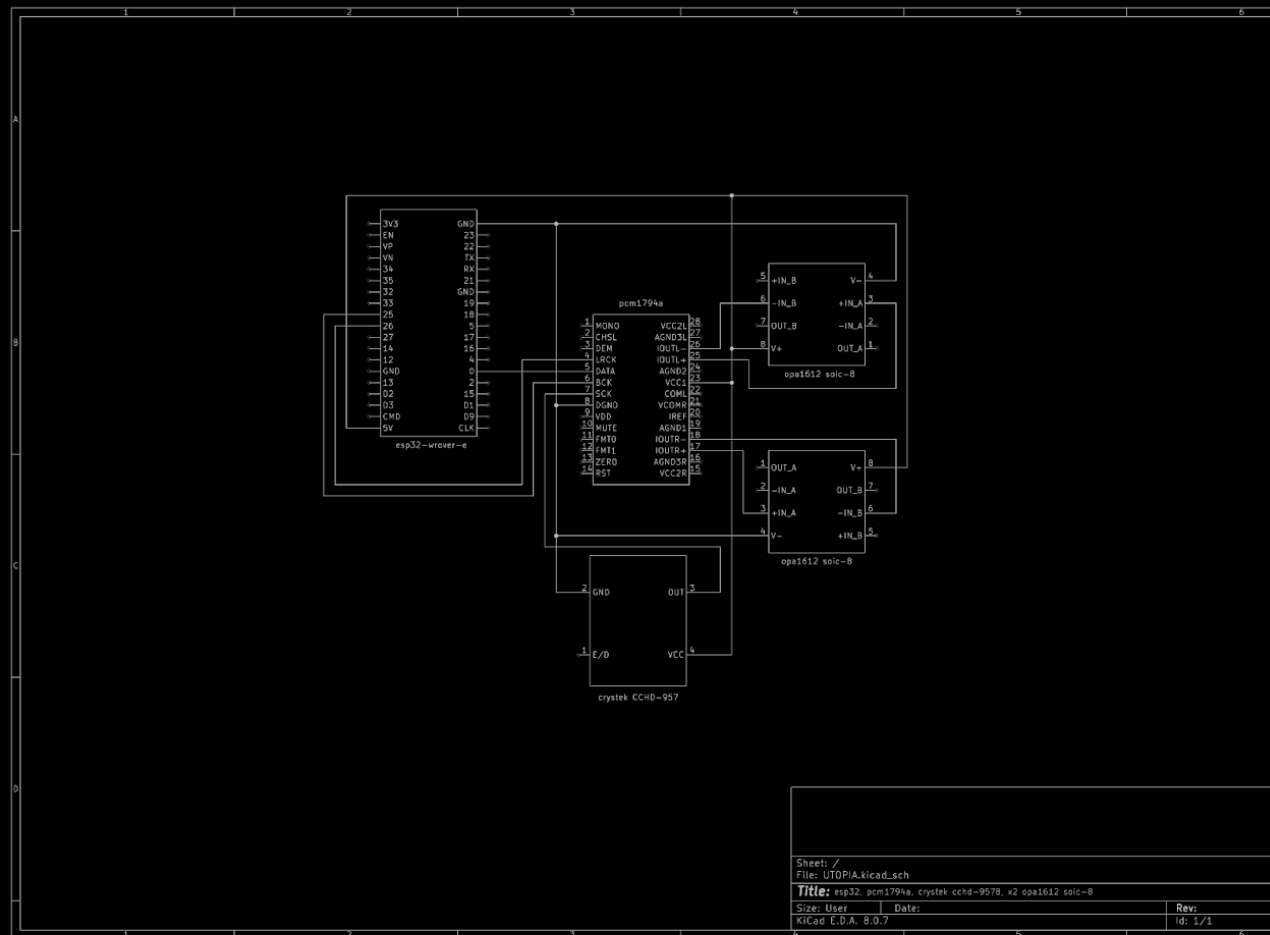
1. internal waveform generation to Speaker output
2. internal audio effect(s)
3. oscillator output
4. microphone input

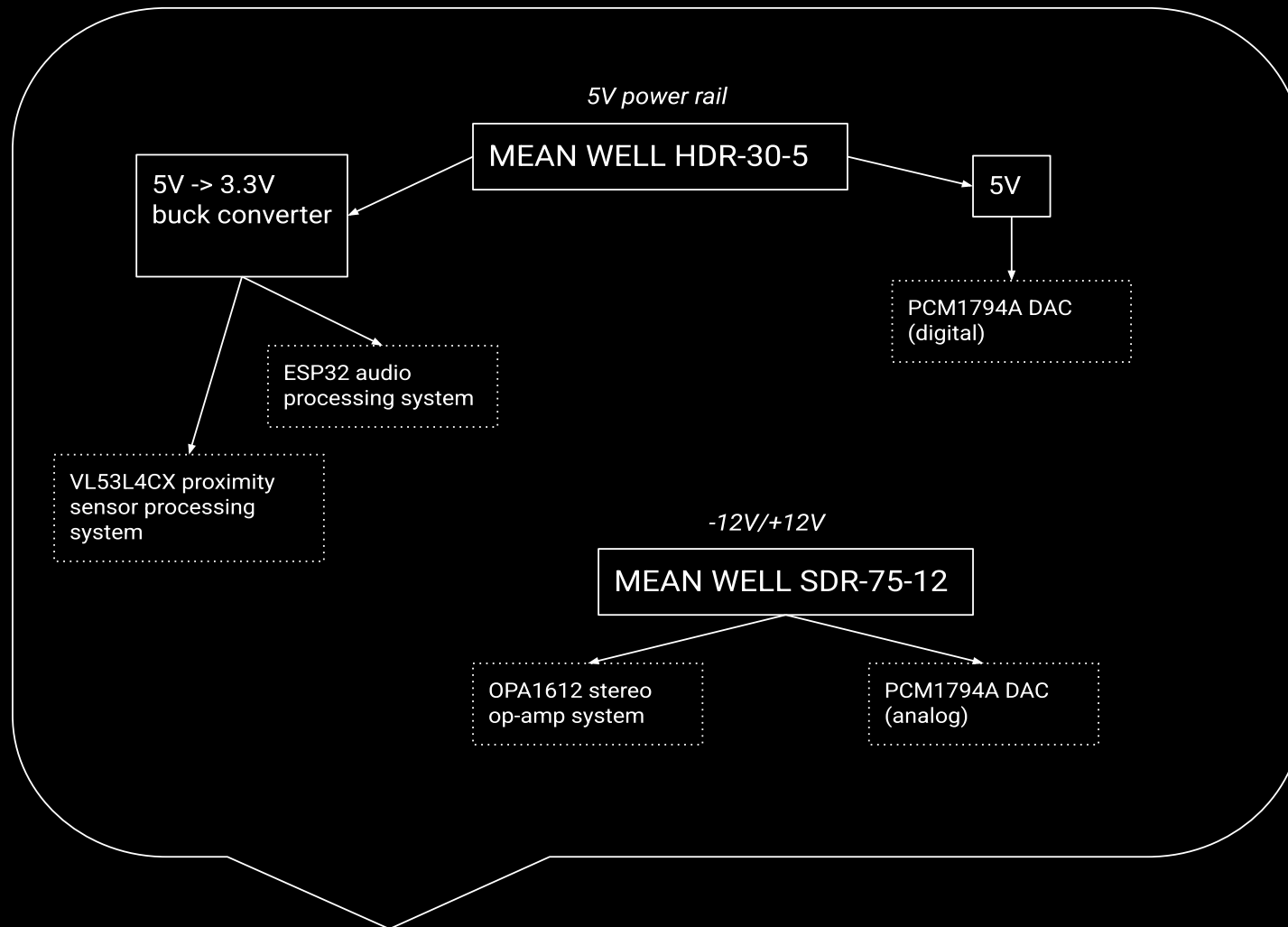
prototype 1 - signal flow





prototype 1 - pinout





prototype 1 - power systems

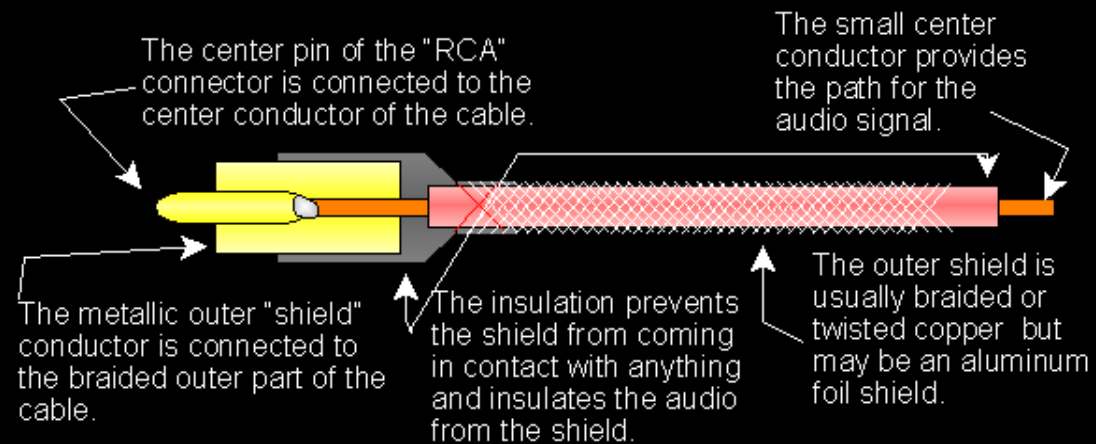
MEAN WELL SDR-75-12 (-12V/+12V dual supply)

[listing](<https://www.mouser.com/ProductDetail/MEAN-WELL/SDR-75-12?qs=hJ2CX3hEdVEsPwk4bX8JoA%3D%3D>)

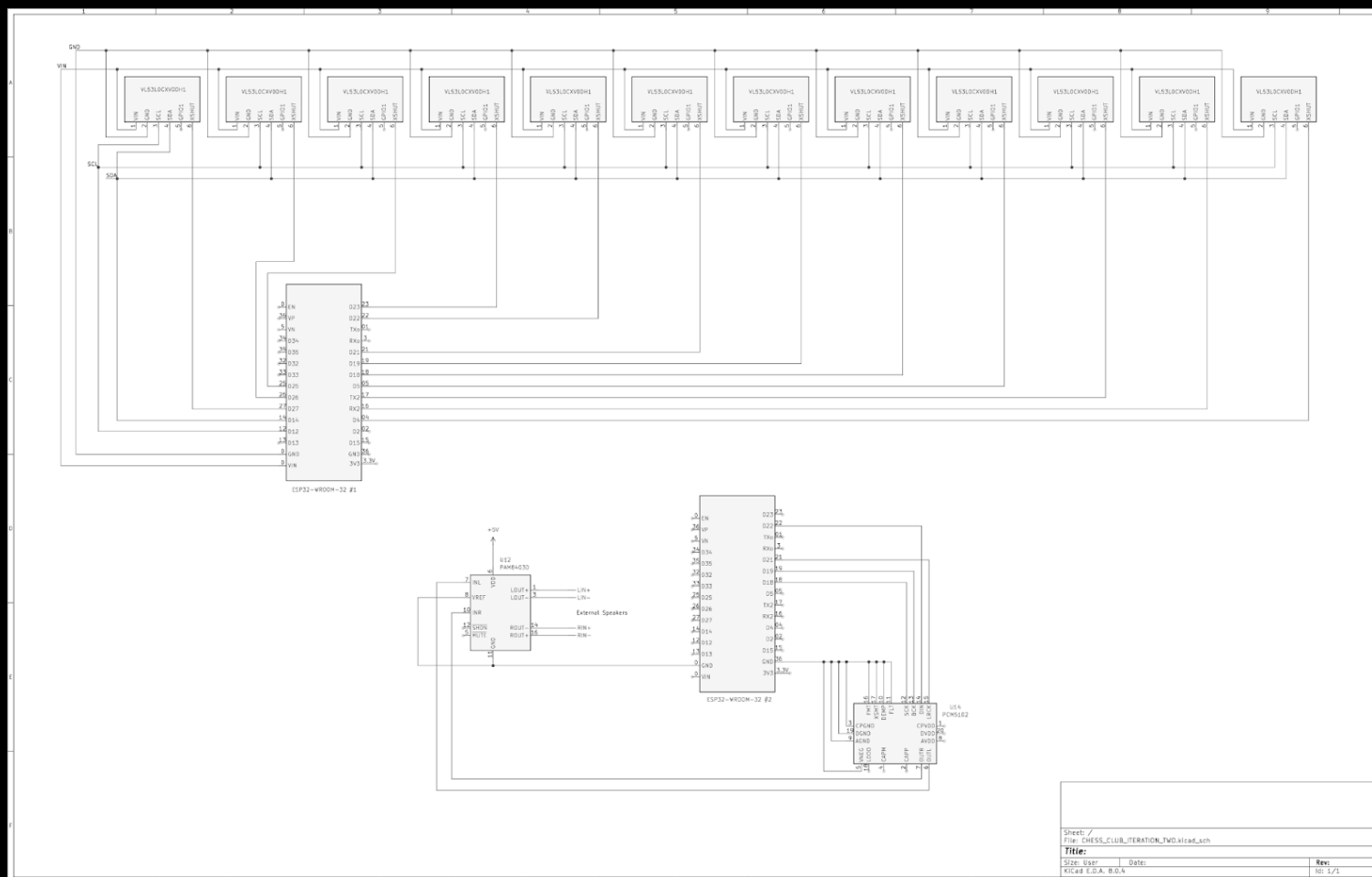
MEAN WELL HDR-30-5 (5V single-line supply)

[listing](<https://www.mouser.com/ProductDetail/MEAN-WELL/HDR-30-5?qs=pHY8AWQbqIMtNGYbObppmw%3D%3D>)

Notes: A transformer may be necessary in addition to shielded RCA cables



ARCHIVE



PRELIMINARY SCHEMATICS SUBMITTED FOR PROPOSAL

Preliminary Schematic Overview:

12 VL53LOCX proximity sensors connected to one ESP32.

PCM5102 DAC > ESP32 > PAM84030 Audio Driver > R/L External Speakers.

Note for modifications in the next version:

2 dedicated ESP32s for processing sensor data

VL53L4CX substitute

PRELIMINARY SCHEMATIC OVERVIEW

** power*

- 5V power supply: MAX98357 amplifier, PCM1794A DAC
- 5V to 3.3V buck converter
- (-12V/+12V) dual power rail

** inputs*

- *sensors*

- VIN to 3.3V
- GND to GND
- SCL to ESP32 SCL
- SDA to ESP32 SDA
- GPIO (none)
- XSHUT to unique GPIO on ESP32

- *microphone*

** processors*

- ESP32s
- amplifiers

- DAC (*Digital-to-Analog Converter*)

*** *outputs***

- *speakers*
- *subwoofer*
- *oled display*